

Cluster 2

To view the latest Map of Science clusters and data, visit the live Map at sciencemap.eto.tech.

Subjects

Research fields: Medicine, Biology, Psychology

The most common research fields of articles in this cluster. [Read more.](#)

Research subfields: Gastroenterology, Radiology, Urology

Common research subfields of articles in this cluster. [Read more.](#)

Key concepts: Gastrointestinal Stromal Tumors, GIST patients, Stromal Tumors, advanced GIST, PDGFRA mutations

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

2.30% of all articles are AI-related

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.98% of English-language articles are related to robotics

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.69% of English-language articles are related to computer vision

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to natural language processing

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to AI safety

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

0.00% of English-language articles are related to cybersecurity

Vital signs

2184 recent articles in the cluster

2184 articles in this cluster were published in the five year period ending in 2023.

Average article is **12.63** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **25.32th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 25.32% of other clusters.

Citation rating: **38.50th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 38.50% of all other articles published in the same year.

80.49% of articles are in English

Not expected to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 1973 of 2184 articles. 1440 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
China	541	0.97
United States	382	2.39
Japan	153	1.95
India	148	0.91
Italy	101	2.90
United Kingdom	93	5.16
Germany	90	5.53
Spain	89	5.39
France	77	6.34
Netherlands	59	7.04

Top funding countries

Data available for 342 of 2184 articles. 318 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
China	171	1.48
United States	84	3.08
Japan	18	6.74
Switzerland	17	7.61
Germany	16	3.23
South Korea	16	1.29
Spain	13	4.96
Italy	5	2.99
Belgium	5	7.02
Norway	5	1.84

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

[Avapritinib in advanced PDGFRA D842V-mutant gastrointestinal stromal tumour \(NAVIGATOR\): a multicentre, open-label, phase 1 trial](#)

2020: The Lancet. Oncology. 221 citations.

[Gastrointestinal stromal tumours](#)

2021: Nature reviews. Disease primers. 270 citations.

[Ripretinib in patients with advanced gastrointestinal stromal tumours \(INVICTUS\): a double-blind, randomised, placebo-controlled, phase 3 trial.](#)

2020: The Lancet. Oncology. 285 citations.

[Survival Outcomes Associated With 3 Years vs 1 Year of Adjuvant Imatinib for Patients With High-Risk Gastrointestinal Stromal Tumors: An Analysis of a Randomized Clinical Trial After 10-Year Follow-up.](#)

2020: JAMA oncology. 131 citations.

[Avapritinib in unresectable or metastatic PDGFRA D842V-mutant gastrointestinal stromal tumours: Long-term efficacy and safety data from the NAVIGATOR phase I trial, 2021](#)

2021: England : 1990). 99 citations.

[The management of metastatic GIST: current standard and investigational therapeutics](#)

2021: Journal of hematology & oncology. 166 citations.

[Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical Practice Guidelines for diagnosis, treatment and follow-up.](#)

2021: Annals of oncology : official journal of the European Society for Medical Oncology. 329 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

[Prognostic indicators for gastrointestinal stromal tumors: a review](#)

2020: Translational oncology. 79 citations.

[A literature review and database of how the primary KIT/PDGFRA variant of a gastrointestinal stromal tumour predicts for sensitivity to imatinib](#)

2022: Cancer genetics. 2 citations.

[Gastrointestinal stromal tumor: a review of current and emerging therapies](#)

2021: Cancer metastasis reviews. 50 citations.

[Clinical characteristics and prognosis of gastrointestinal stromal tumors with rare site metastasis \(Review\)](#)

2022: Oncology letters. 11 citations.

[The GIST of Advances in Treatment of Advanced Gastrointestinal Stromal Tumor.](#)

2022: American Society of Clinical Oncology educational book. American Society of Clinical Oncology. Annual Meeting. 42 citations.

[Management of Gastrointestinal Stromal Tumors: An Update for Surgeons](#)

2024: Current problems in surgery. 0 citations.

[2023 GEIS Guidelines for gastrointestinal stromal tumors](#)

2023: Therapeutic advances in medical oncology. 32 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Avapritinib in advanced PDGFRA D842V-mutant gastrointestinal stromal tumour \(NAVIGATOR\): a multicentre, open-label, phase 1 trial](#)

2020: The Lancet. Oncology. 221 citations.

[Prognostic indicators for gastrointestinal stromal tumors: a review](#)

2020: Translational oncology. 79 citations.

[Gastrointestinal stromal tumours](#)

2021: Nature reviews. Disease primers. 270 citations.

[Ripretinib in patients with advanced gastrointestinal stromal tumours \(INVICTUS\): a double-blind, randomised, placebo-controlled, phase 3 trial.](#)

2020: The Lancet. Oncology. 285 citations.

[Gastrointestinal stromal tumours](#)

2021. 271 citations.

[Survival Outcomes Associated With 3 Years vs 1 Year of Adjuvant Imatinib for Patients With High-Risk Gastrointestinal Stromal Tumors: An Analysis of a Randomized Clinical Trial After 10-Year Follow-up.](#)

2020: JAMA oncology. 131 citations.

[Avapritinib in unresectable or metastatic PDGFRA D842V-mutant gastrointestinal stromal tumours: Long-term efficacy and safety data from the NAVIGATOR phase I trial, 2021](#)

2021: England : 1990). 99 citations.

[The management of metastatic GIST: current standard and investigational therapeutics](#)

2021: Journal of hematology & oncology. 166 citations.

[Gastrointestinal stromal tumours: ESMO-EURACAN-GENTURIS Clinical Practice Guidelines for diagnosis, treatment and follow-up.](#)

2021: Annals of oncology : official journal of the European Society for Medical Oncology. 329 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 2171 out of 2184 articles. 405 of these articles came from these sources.

Source	Articles (last five years)
Cureus	97
Research Square (Research Square)	61
Reactions weekly	49
Frontiers in oncology	42
Cancers	35
International journal of surgery case reports	34
Journal of Clinical Oncology	24
中华胃肠外科杂志	22
Annals of surgical oncology	22
Clinical cancer research : an official journal of the American Association for Cancer Research	19

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 2106 of 2184 articles. 175 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Michael C. Heinrich	52	24.29
Robin L. Jones	52	37.54
Sebastian Bauer	47	22.33
Hans Gelderblom	45	17.62
César Serrano	44	12.84
Suzanne George	41	22.47
Margaret von Mehren	40	20.49
Patrick Schöffski	36	13.68
Jean-Yves Blay	35	17.77
Piotr Rutkowski	31	19.95

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 2001 of 2184 articles. 227 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
Sun Yat-sen University - China	60	1.2
Dana-Farber Cancer Institute - United States	53	6.35
Institute of Cancer Research - United Kingdom	46	8.55
Memorial Sloan Kettering Cancer Center - United States	44	7.45
Leiden University Medical Center - Netherlands	43	7.24
Fox Chase Cancer Center - United States	43	8.71
Centre Léon Bérard - France	41	10.73
The First Affiliated Hospital, Sun Yat-sen University - China	38	1.36
Peking University - China	37	1.55
Portland VA Medical Center - United States	35	7.39

Organization types

This table covers the 1856 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 2184 articles total). [Read more](#)

Commercial	Education	Nonprofit	Government
1.40%	25.97%	35.88%	1.02%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 111 out of 2184 articles. 60 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Deciphera Pharmaceuticals (United States)	24	6.22
Blueprint Medicines (United States)	11	10.74
Molecular Oncology (United States)	4	5.33
Springer Nature (New Zealand)	4	7.4
United Imaging Healthcare (China)	4	6.01
AstraZeneca (United States)	3	2.98
Medica (Italy)	3	1.22
Novartis (Switzerland)	3	2.67
Siemens (China)	3	2.23
AstraZeneca (United Kingdom)	3	2.64

Top funders

Data available for 520 out of 2184 recent articles. 195 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Natural Science Foundation of China	95	1.71
National Institutes of Health - United States	43	3.33
Department of Veterans Affairs - United States	22	4.91
Deciphera Pharmaceuticals, LLC	17	5.60
Novartis	16	2.83
Novartis (Switzerland)	16	2.94
Zhejiang Province - China	14	1.84
GIST Cancer Research Fund	14	5.35
Pfizer	14	1.81
Deciphera Pharmaceuticals (United States)	12	3.13

Funder types

This table covers the 287 articles in this cluster from the last five years with data about their funding organization types (out of 2184 articles total).

Commercial	Education	Nonprofit	Government
19.03%	9.65%	14.21%	57.10%

Patents and industry

494 patents cite articles in this cluster (**96.91th** percentile among all clusters)

2.91% of articles in the cluster are cited by patents

1.3% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
SIGNATURES OF CLINICAL OUTCOME IN GASTRO INTESTINAL STROMAL TUMORS AND METHOD OF TREATMENT OF GASTROINTESTINAL STROMAL TUMORS	28
2-(3-AMINOARYL)AMINO-4-ARYL-THIAZOLES AND THEIR USE AS C-KIT INHIBITORS	25
AMORPHOUS KINASE INHIBITOR FORMULATIONS AND METHODS OF USE THEREOF	22
COMPOSITIONS COMPRISING PHOSPHODIESTERASE INHIBITORS FOR USE IN THE TREATMENT OF A SOLID TUMOR IN A HUMAN PATIENT	22
COMBINATION THERAPY FOR THE TREATMENT OF MASTOCYTOSIS	21
COMPOSITIONS OF 1-(4-BROMO-5-(1-ETHYL-7-(METHYLAMINO)-2-PENDANT OXY-1, 2-DIHYDRO-1, 6-NAPHTHYRIDIN-3-YL)-2-FLUOROPHENYL)-3-PHENYLUREA	21
COMBINATION THERAPY FOR TREATING GASTROINTESTINAL STROMAL TUMORS	21
REPRETINIB FOR TREATMENT OF GASTROINTESTINAL STROMAL TUMORS	21
METHODS OF TREATING GASTROINTESTINAL STROMAL TUMORS	20

Top patent assignees

Patent assignee	Same-cluster citations
INNVELOP LIMITED	49
NOVARTIS (SWITZERLAND)	45
DECIPHERA PHARMACEUTICALS LLC	30
INST BERGONIÉ	30
UNIVERSITÉ DE BORDEAUX	30
INST BERGONIE	29
CHIBON FREDERIC	28
COINDRE JEAN-MICHEL	28
INSTITUT BERGONIE	28

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
11263	303	gastrointestinal stromal tumors, endoscopic full-thickness resection, gastric GISTs, stromal tumors, gastric subepithelial lesions
22147	39	Therapeutic drug monitoring, Tyrosine kinase inhibitors, TDM, imatinib plasma concentrations, human plasma
1516	33	Radiomic features, deep learning radiomics, cell lung cancer, MRI Radiomics Model, breast cancer
20051	29	case report, Primary gastric leiomyosarcoma, LMS, tumor, gastrointestinal stromal tumors
738	29	Pheochromocytoma and Paraganglioma, Paragangliomas, Metastatic PPGLs, Neuroendocrine Tumors, SDHB PPGLs
6888	24	stem cell factor, tyrosine kinase KIT, KIT gene, receptor tyrosine kinase, KIT mutations
43657	22	Gastrointestinal stromal tumors, collision tumor, gastric GIST, GIST patients, stromal tumors
1309	19	soft tissue sarcoma, extremity soft tissue, sarcoma patients, primary extremity STS, tissue sarcomas results
3715	17	circulating tumor DNA, tumor DNA, cell-free DNA, tumor DNA analysis, ctDNA MRD detection
66845	16	cancer cells, cell proliferation, Cell Lung Cancer, cell apoptosis, long non-coding RNA

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
11263	74	gastrointestinal stromal tumors, endoscopic full-thickness resection, gastric GISTs, stromal tumors, gastric subepithelial lesions
1516	52	Radiomic features, deep learning radiomics, cell lung cancer, MRI Radiomics Model, breast cancer
738	26	Pheochromocytoma and Paraganglioma, Paragangliomas, Metastatic PPGLs, Neuroendocrine Tumors, SDHB PPGLs
6888	23	stem cell factor, tyrosine kinase KIT, KIT gene, receptor tyrosine kinase, KIT mutations
5998	19	systematic reviews, Preferred Reporting Items, reporting systematic, JAMA Neurology JAMA, JAMA JAMA Network
44955	19	NTRK gene fusions, TRK Fusion Cancer, NTRK fusion-positive solid, TRK inhibitors, tropomyosin receptor kinase
3715	19	circulating tumor DNA, tumor DNA, cell-free DNA, tumor DNA analysis, ctDNA MRD detection
22147	14	Therapeutic drug monitoring, Tyrosine kinase inhibitors, TDM, imatinib plasma concentrations, human plasma
429	14	neurofibromatosis type, Plexiform Neurofibromas, tumor, Selumetinib, Ras
19	12	chronic myeloid leukemia, Research Funding, CML patients, chronic phase CML, tyrosine kinase inhibitors

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
cran/iai	1	1
alexdobin/STAR	1	1851
AIM-Harvard/pyradiomics	1	1159
mskcc/mutation-signatures	1	60
yufuowkin/GIST	1	2
lasseignelab/CINmetrics	1	7